Speech Emotion Recognition

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ABSTRACT

Speech Emotion Recognition (SER) stands for one of the important technologies in the development of human-computer interaction, which allows the machine to detect human emotion through speech. This paper has presented a broad methodology for the development of an efficient enough SER system using deep learning techniques. In this regard, we combined four different emotional speech datasets: CREMA-D, RAVDESS, SAVEE, and TESS, to construct a broad dataset that covers almost all the categories of emotions. In order to handle issues of data imbalance and limited diversity, we employed techniques of data augmentation such as noise injection, time stretching, and pitch shifting. Further balancing of this dataset is done using the SMOTE method.

Feature extraction methodologies ranged from traditional audio features like Zero Crossing Rate and Mel-Frequency Cepstral Coefficients to some of the interesting advanced deep-learning-based features, namely Emotion2Vec and Wav2Vec 2.0.

Various deep learning models were evaluated, including Convolutional Neural Networks (CNN), Long Short-Term Memory networks (LSTM), and Deep Neural Networks (DNN). The best performance is shown by using Emotion2Vec features combined with a DNN architecture. It achieves 80.5% on validation accuracy and 78.5% on test accuracy, which indicates the effectiveness in utilizing specialized deep learning features designed for emotion recognition tasks. The developed SER system has huge potential to improve human-computer interaction in diverse applications. However, the model is sensitive to recording quality and the intensity of the emotional expression when tested on non-professional self-recorded audio, which is a limitation. Future work will be related to enhancing the generalization capability of the model for real-world conditions by the usage of more diversified and spontaneous speech samples, optimization of computational efficiency, and exploring multimodal emotion recognition techniques.

Keywords: Speech Emotion Recognition, Deep Learning, Emotion2Vec, Data Augmentation, Feature Extraction, Deep Neural Networks

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1. Introduction

1.1 Problem Context

Speech Emotion Recognition is a relatively new field of study these days within human-machine interaction to recognize one's emotions and affective states from speech signals. The motivation behind this project stems from the inherent challenges and potential of SER in enhancing the interaction between humans and machines. Emotion recognition either through their faces or by traditional means is a rather technically complicated and financially costly procedure, as it involves high-resolution cameras and complex processing. By contrast, vocal emotions are much more accessible and inexpensive as a medium for identifying one's emotions. Therefore, this paper aims at enhancing the accuracy and efficiency of the SER using deep learning techniques.

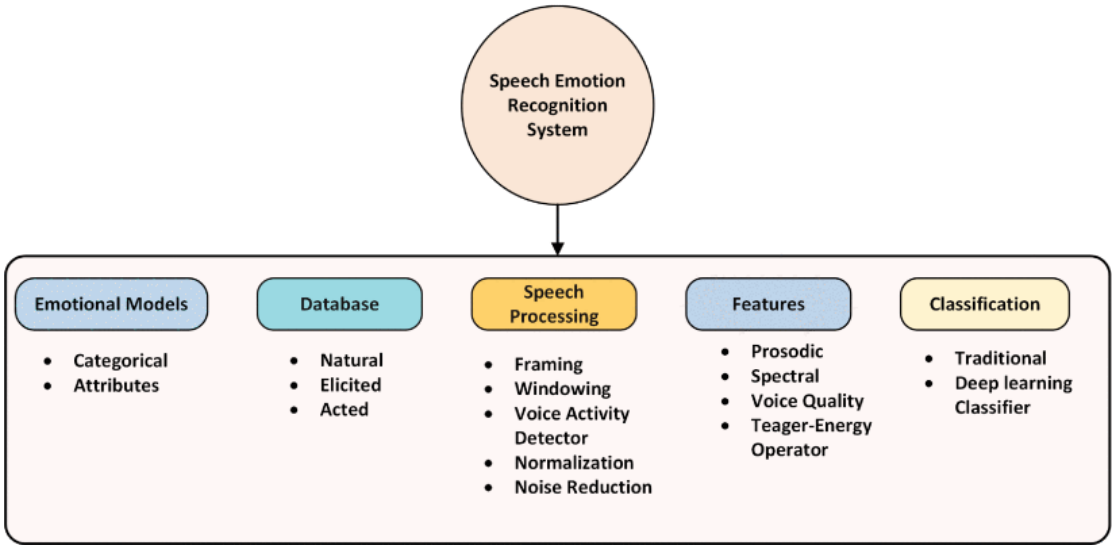


FIGURE 1. General SER System.

1.2 Data Preparation

We used several datasets, each containing audio recordings of actors uttering a variety of emotions. Each of these forms foundational data for the training and testing of our machine learning models in emotion recognition. To facilitate ease of processing and training, this data was all brought into one framework. Details on each dataset are as follows:

- Crowd Sourced Emotional Multimodal Actors Dataset (CREMA-D):
  It comprises 7,442 audio clips from 91 actors, of which 48 are male and 43 are female, aged between 20 and 74. The actors are of several racial and ethnic groups, including African American, Asian, Caucasian, Hispanic, and others. Each of these actors has performed 12 sentences, delivered in one of six emotions-anger, disgust, fear, happiness, neutral, and sadness-each of these with four emotion intensities: low, medium, high, and unspecified.

- Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS):
  The RAVDESS consists of 1,440 audio files performed by 24 professional actors uttering the same two phonetically matched sentences in a neutral North American accent. Data are recorded for eight named emotions - calmness, happiness, sadness, anger, fear, surprise, disgust, and neutral - in two levels of intensity, normal and strong, and in a neutral tone.
- Surrey Audio-Visual Expressed Emotion (SAVEE):
  This data set was recorded by four male English speakers aged between 27 and 31, capturing seven emotions: anger, disgust, fear, happiness, sadness, surprise, and neutral. Every speaker spoke 15 TIMIT sentences per emotion with three shared sentences, two emotional sentences unique to each emotion, and ten general sentences. A total number of recordings was 120 from each speaker.
- Toronto Emotional Speech Set (TESS):
  Of these, TESS consists of 2,800 recordings by two female speakers aged 26 and 64. Each file expresses one of seven emotions: anger, disgust, fear, happiness, pleasant surprise, sadness, and neutral. In each recording, 200 unique target words are spoken within the carrier phrase, “Say the word _”.

1.3 Motivation

The motivation for developing an SER system is multifaceted. First, human communication is deeply embedded with emotional cues, which are conveyed through both linguistic and paralinguistic features of speech. These acknowledged emotions will go a long way in smoothing human-machine interaction. Imagine SER put into place at call centers that classify calls into classes of emotional states of customers, thereby allowing for the determination of unsatisfied customers. Conversely, these are done in order for their companies to work on improving their services. In automotive systems, it does detect the mental state of drivers to prevent accidents by taking safety measures when detecting stress or anger.

While being of importance, in the case of traditional SER systems, several obstacles exist. They have high dependence on handcrafted features, with poor generalization across different datasets or languages. Moreover, there is limited diversity in datasets and emotional categories are imbalanced, which further hampers the accuracy and robustness of a model. The current deep learning techniques are prone to issues involving overfitting, computational inefficiency, or sub-optima feature representation.

This project thus focused on the development of such gaps using cutting-edge deep learning architectures and techniques such as data augmentation. We studied hybrid models by integrating several datasets, hence achieving better accuracy, robustness, and generalized performance of speech emotion recognition systems.

1.4 Problem Statement

The primary problem this project addresses is the lack of robust and accurate SER systems capable of generalizing across diverse datasets and emotional categories.

Specifically, we aim to develop a deep learning-based SER system that leverages diverse datasets, advanced data preprocessing, and novel model architectures to improve classification accuracy and generalization.

1.5 Conclusion

In summary, this project aims to solve technical and practical challenges with a deep learning-based classifier for speech emotion recognition. In this project, diverse and comprehensive datasets will be utilized in the design of a system that precisely recognizes the emotional states from speech to further improve human-machine interaction in various applications. Successfully implemented, manifold industries may be influenced, the user experience enhanced, and participation in continuous research related to affective computing actively taken.

2. Related Work

It has been over two decades since speech emotion recognition began its harvest of studies. Early works applied traditional machine learning methods with handcrafted feature extraction to classify speech records into one of a few classes of emotions. In recent years, major development has been seen in deep learning, which revolutionized the field by enabling automatic feature extraction and improving classification accuracy. This chapter will review relevant literature, datasets, methodologies, and highlight the gaps that this study addressed.

2.1 Traditional Approaches to SER

- Traditional SER methods rely on machine learning algorithms that include: Support Vector Machines (SVMs): Effective in handling high-dimensional features but limited in capturing temporal patterns of audio signals.
- Hidden Markov Models (HMMs): Suitable for modeling sequential data but often struggle with complex emotional cues.
- Decision Trees: Decision trees are easy to deploy, but suffer from overfitting and poor scalability.

While these methods laid a foundation for SER, they are mostly based on handcrafted features and generalize poorly to practical scenarios.

2.2 Deep Learning Approaches to SER

Deep learning has revolutionized SER with models that learn features directly from raw audio data. Prominent architectures include:

2.2.1 Multilayer Perceptron (MLP)

Alnuaim et al. (2022) investigated the use of MLP classifiers for speech emotion recognition in human-computer interaction applications. Concretely, their work described that MLP is capable of representing complex patterns in speech signals to enable real-time detection of emotional states of individuals. In an MLP architecture, a multilayer neural network configuration is considered wherein processing neurons at every layer will, in turn, build higher-order hierarchical representations that help classify small changes in emotions from speech.

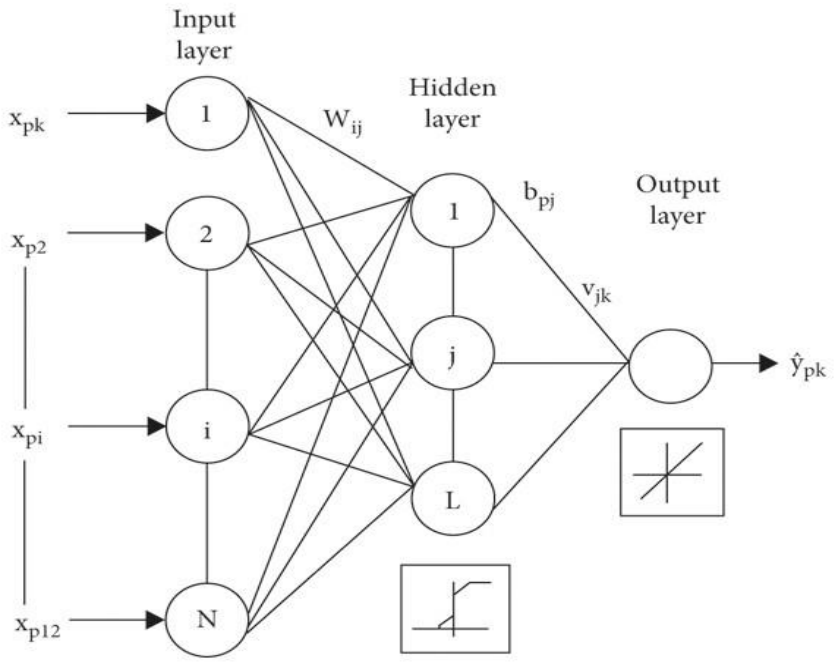


FIGURE 2. The basic structure of a multilayered perceptron.

2.2.2 Convolutional Neural Networks (CNNs)

Building on the neural architecture models, Alnuaim et al. (2022) studied CNNs for speaker emotion detection. It is expected that CNNs will promote robust feature extraction in emotion classification, as they tap the spatial dependencies in spectrograms. While there is a connected problem of computational demands and overfitting issues with CNNs, they do come with enormous advantages in terms of processing audio data efficiently.

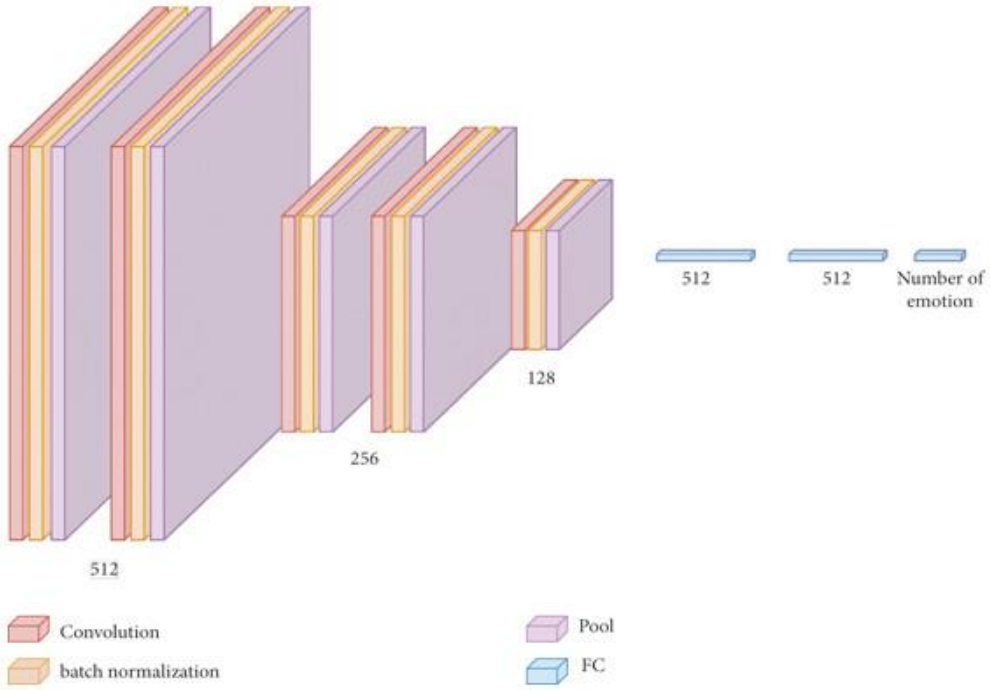


FIGURE 3. 1D CNN architecture.

- Recurrent Neural Networks (RNNs): Capture temporal dependencies in audio signals, making them suitable for sequential data.
- Transformers: Leverage self-attention mechanisms to model global dependencies, offering superior performance in tasks requiring context awareness.

2.3 Comprehensive Approaches to SER

Wani et al. (2021) conducted a review on speech emotion recognition systems with the use of diverse methodologies, from MLPs to CNNs and hybrid models. In their review they pointed out that a shift is taking place toward the use of multiple modalities in audio and text data to improve the recognition rate of emotions in diverse linguistic contexts. They emphasized the great improvements deep learning has brought to SER. While CNNs have been relatively more successful among these in speech signal processing, in particular, when applied to the spectrogram of the speech. The automatically learned hierarchical feature representations by CNNs have emerged to show more potential to identify complex emotional patterns from raw audio signals. RNNs - and especially Long Short-Term Memory (LSTM) networks - are perfectly suited to extract temporal dependencies within speech and, thus, for sequential data such as audio signals. The combination of CNNs and RNNs, often referred to as hybrid models, leverages the strengths of both architectures, providing robust feature extraction and temporal modeling capabilities.

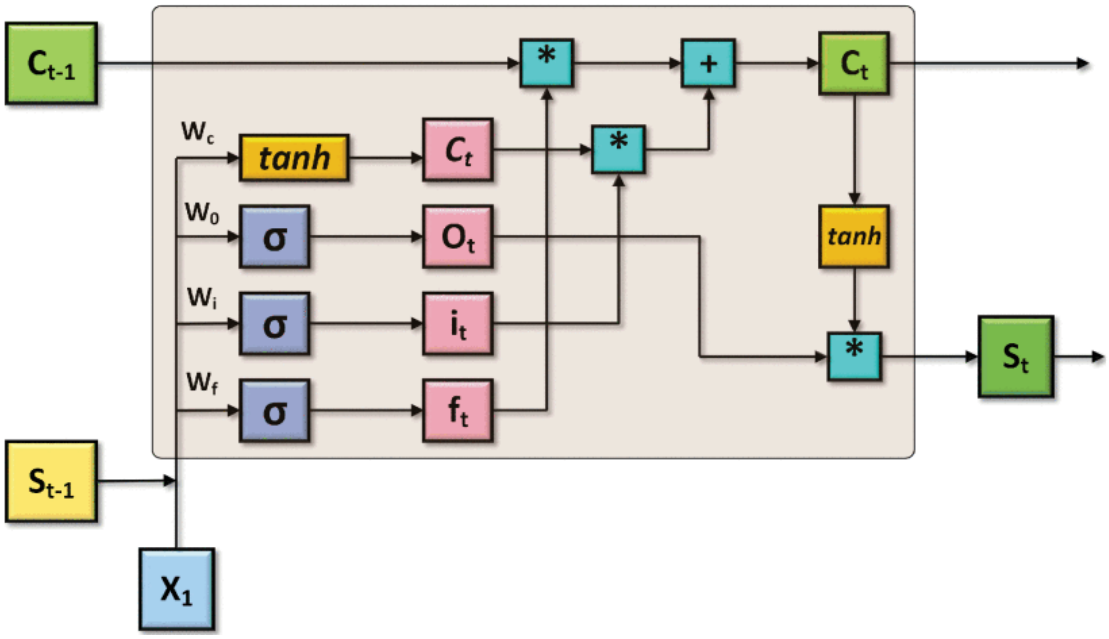


FIGURE 4. Basic Architecture of LSTM.

2.4 Critique of Existing Work

Pros:

- Advances in deep learning have significantly improved accuracy and robustness in SER systems.
- Hybrid models effectively combine the strengths of CNNs and RNNs.
- Data augmentation techniques enhance the generalization of models to unseen data.

Cons:

- Many studies rely on small and imbalanced datasets, limiting the generalization of models.
- Computational demands of deep learning models hinder their practical deployment.
- Few works explore cross-dataset generalization or multilingual SER.

2.5 Research Gaps

This study addresses several key gaps in existing research:

- Integration of diverse datasets to improve model generalization.
- Use of advanced data augmentation and feature extraction techniques to mitigate class imbalance.
- Exploration of hybrid deep learning models for optimal performance.

3. Research Methodology

3.1 Data Processing

3.1.1 Data Augmentation

Data augmentation is a method of artificially increasing the diversity of the training datasets when one has limited available data. This technique becomes very important in any machine learning project since it helps in improving the model's generalization power and performance. By introducing various transformations to the original data, new and diverse samples are generated. In our project, we used some augmentation techniques specifically for audio datasets, including Noise Injection, Time Stretching, and Pitch Shifting.

a. Noise Injection: Noise is added to the audio signal to simulate different noisy environments. The formula for noise injection is:

$$y_{noise} = y + \alpha n$$

where y is the original audio signal, n is the noise signal, and α is the noise level.

b. Time Stretching: Time stretching changes the speed of the audio signal without altering its pitch. This is done using the following transformation:

$$y_{stretched} = stretch(y, \beta)$$

where β is the stretching factor.

c. Pitch Shifting: Pitch shifting changes the pitch of the audio signal without affecting its duration. The transformation is:

$$y_{pitched} = pitch[y + \gamma]$$

where γ is the pitch shift amount.

3.1.2 Feature Extraction

Feature extraction plays a vital role in analyzing and identifying relationships among various elements. Since audio data is not inherently interpretable by models, it must be transformed into a format through feature extraction that machines can process. An audio signal is inherently three-dimensional, with its dimensions corresponding to time, amplitude, and frequency.

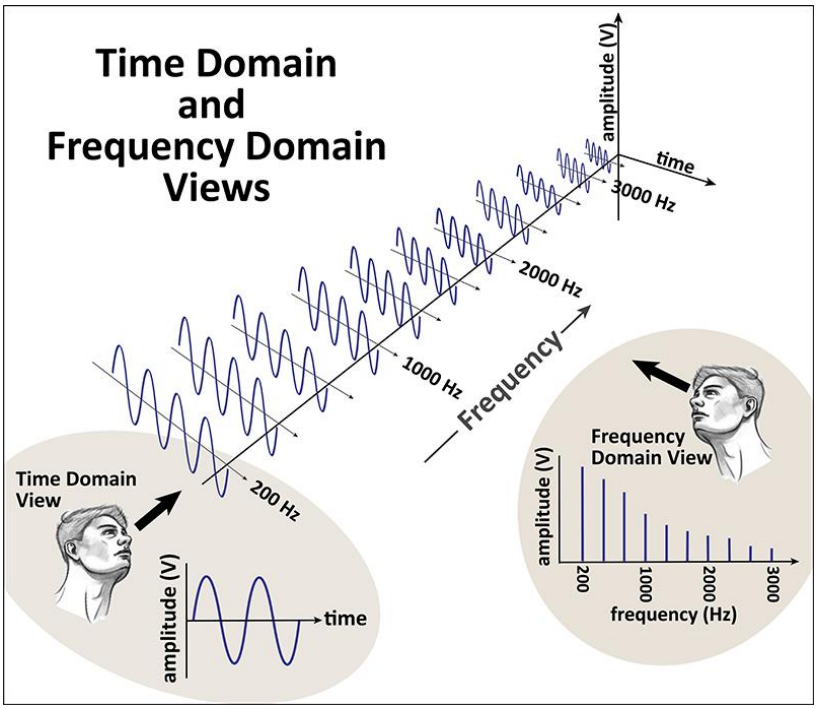


FIGURE 5. Audio signals for 3D signals

We focused on extracting features that capture different aspects of the audio signal, including:

Traditional audio features (manually designed features)

1. Zero Crossing Rate (ZCR): The zero crossing rate measures how often the audio signal crosses the zero amplitude line, switching between positive and negative values. It serves as an indicator of the signal's texture or level of noisiness.

The ZCRs are computed as follows:

$$ZCR = \frac{1}{N-1}\sum_{n=1}^{N-1} 1_{\{sign(y[n]) \neq sign(y[n-1])\}}$$

where N is the number of samples, and 1 is the indicator function that is 1 if the condition is true and 0 otherwise.

2. Chroma Short-Time Fourier Transform (Chroma_stft): The chroma short-time Fourier transform represents the energy distribution across different pitches in the audio signal. It captures the harmonic and melodic content of the audio.

The Chroma feature is computed as follows:

$$Chroma_i(t) = \sum_{k \in K_i} |X_k(t)|$$

where K_i is the set of frequency bins corresponding to the i-th pitch class, and $|X_k(t)|$ is the magnitude of the Fourier transform at frame t and frequency bin k.

3. Mel-Frequency Cepstral Coefficients (MFCC): MFCCs are used to represent the short-term power spectral of an audio signal. This is done by mapping the logarithm of the power spectrum onto a nonlinear mel scale and then applying a linear cosine transform to this representation.

The MFCCs are computed as follows:

$$MFCC_n = \sum_{m=1}^{M} log(S_m)\, cos\left(\frac{n\pi(m-0.5)}{M}\right)$$

where S_m is the mel-scaled power spectrum, M is the number of mel bands, and n is the number of cepstral coefficients.

4. Root Mean Square Value: The RMS value quantifies the energy or loudness of an audio signal. It squares each of the values of the signal, after which it computes an average of these squares and takes the square root of that average.

The RMS value is computed as follows:

$$RMS = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

where N is the number of samples and y_n is the amplitude of the n-th sample.

5. Mel Spectogram: A Mel Spectrogram is a time-frequency representation of the audio signal. It provides a detailed view of how the spectral content of the signal varies with time.

These features will be used to train our model.

The Mel Spectrogram is computed as follows:

$$S(m,t) = \sum_{f=0}^{F-1} |X(f,t)|^2 \cdot h_m(f)$$

where $|X(f,t)|^2$ is the power spectrum of the audio signal at frequency f and time t, and $h_m(f)$ is the mel filter bank.

Deep learning features (data-driven features)

1. Emotion2Vec: Emotion2Vec is a feature extraction method that designed specifically for emotion recognition tasks, which has several significant advantages in speech emotion recognition tasks.
   Core Architecture:

- Input feature processing: Input audio signal $X \in RT$

  Map waveform signals into short-term features using models such as CNN or other front-end models Z:

$$Z = FeatureExtractor(X)$$

- Emotional Representation Learning: Use Transformer model to model Z and generate contextual representation C at the time step.

$$C = Transformer(Z)$$

- Feature pooling: In order to obtain global emotional features of audio, Emotion2Vec may use global pooling methods (such as mean pooling or attention mechanisms).

$$F = Pool(C)$$

Among them, $F \in Rd$ is the final extracted emotion vector.

Technical features:

- Specially designed for emotional modeling: Compared to general speech feature extraction tools such as MFCC, Log mel, Spectrogram, it captures features that are more in line with emotional semantics. Emotion2Vec has been optimized in terms of the emotional characteristics of speech signals, and can extract deep semantic features that reflect emotional changes.
- Using deep learning driven feature learning: It adopts pre trained deep learning models, and the emotion embeddings learned from massive speech data can capture complex nonlinear patterns (Ma et al., 2023). This deep learning based feature extraction far exceeds the expressive power of manually designed features such as MFCC, Chroma, Pitch, etc.
- High dimensional dense features: The extracted Emotion2Vec features are high-dimensional dense features that contain rich speech emotion signals and are robust to noise and interference. These high-dimensional features can represent subtle differences between different emotions in more detail.
- End to end optimization: The Emotion2Vec model undergoes end-to-end optimization, from raw audio input to emotion embedding generation, and the entire process is continuous and collaborative optimization. This enables it to automatically adapt to a variety of different emotional data and language environments.

By utilizing Emotion2Vec, the performance of the emotion classification model can be significantly improved. Its advantages are mainly reflected in:
Higher classification accuracy;
Faster model training;
Cross lingual adaptability;
Ability to model complex emotional patterns.
In addition, Emotion2Vec has the potential for continuous updates and may become the mainstream standard in more emotion recognition tasks in the future.

2. Wav2vec 2.0: Wav2vec 2.0 is an audio representation learning model proposed by Meta (formerly Facebook), aimed at extracting high-quality features from raw audio waveforms in an unsupervised manner. Its core idea draws on pre training and self supervised learning methods in natural language

processing, pre training with large-scale unlabeled audio data and fine-tuning with a small amount of labeled data to achieve a wide range of downstream task applications (Schneider et al., 2019).

Core Architecture:

- Feature extractor (CNN): Input the original waveform $X \in RT$ and an audio signal of length T.
  The convolutional feature extractor converts the original waveform into a shorter feature sequence $Z \in RT' \times d$, with the formula:
  $$Z = CNN(X)$$
  Among them, T′ is the result of shortening the time step, and d is the feature dimension.
- Discrete representation learning: Discretize feature Z using a quantization module to obtain feature representation Q, with the formula:
  $$Q = Quantize(Z)$$
  (For comparing learning objectives)
- Context representation (based on Transformer): Using Transformer model to map feature sequence Z to context representation C:
  $$C = Transformer(Z)$$
  Among them, $C \in RT' \times d$.
- Loss function (contrastive learning): Through contrastive learning, the model matches the correct contextual representation C with the discrete representation Q:
  $$L = -\sum_{t \in T} \quad log\ (\frac{\sum_{Q' \in Q} \quad exp\ (sim(C_t, Q'))}{exp\ (sim(C_t, Q_t))})$$
  Among them, $sim(Ct, Qt)$ is the cosine similarity and Q is the negative sample set.

Technical features:

- Self supervised learning: wav2vec 2.0 adopts a self supervised learning framework based on contrastive learning (Baevski et al., 2020). The model learns latent representations from unlabeled audio during the pre training stage through mask prediction strategy. Its loss function encourages the model to compare temporal features and capture local and global temporal dependencies.
- End to end modeling: Unlike traditional audio processing that relies on manually designed features such as MFCC or Spectrogram, wav2vec 2.0 directly operates on the original waveform, extracting features that better reflect the true content and speech information of the audio.
- Multi layer Transformer architecture: Its feature extractor includes a Convolutional Neural Network (CNN) for preliminary extraction of low-level features, combined with Transformer layers to capture contextual information over long time spans in audio.

3.1.3 Sample Balance

Synthetic Minority Over-sampling Technique (SMOTE) was used to balance the imbalanced data. It increases its quantity by generating synthetic samples between minority class samples, rather than simply copying existing samples. This can more effectively expand the minority class distribution and avoid overfitting of the model due to duplicate data.

Steps of SMOTE:

1. Identify Minority Samples: SMOTE first identifies the minority class samples in the dataset.
2. Find Nearest Neighbors: For each minority sample x, SMOTE computes its $k-$ nearest neighbors within the same class using a distance metric (e.g., Euclidean distance).
3. Generate Synthetic Samples: SMOTE randomly selects one of the nearest neighbors $xneighbor$ and generates a new synthetic sample $xnew$ as follows:

$$xnew = x + \lambda \cdot (xneighbor - x)$$

Here, λ is a random number between 0 and 1, ensuring that the new sample lies on the line segment between x and $xneighbor$.

By using SMOTE, the dataset in the code is balanced, with each class having an equal number of training samples, leading to a fairer and more effective training process.

3.1.4 Other Procedures

To ensure the machine learning model can effectively process the audio data, we implemented the following key preprocessing steps:

1. Label Encoding: Initially, categorical emotion labels were converted into numerical form using a label encoding technique. This step mapped each emotion category to a unique integer, facilitating the subsequent application of machine learning algorithms.
2. One-Hot Encoding: After label encoding, the categorical emotion labels were converted into a binary matrix representation using one-hot encoding. This step allowed the model to provide each category of emotion independently to help the model differentiate properly between emotions.
3. Standardization: This is done to ensure homogeneity among the features so that no one feature can dominate the entire training process of the model. Lastly, for scaling the audio features, the following formula was used:

$$z = \frac{x - \mu}{\sigma}$$

4. where x is the original feature value μ is the mean of the feature σ is the standard deviation of the feature.
5. Spanning Input Dimension: Unless the architecture of a neural network allows for that, data dimension adjustments had to be made. Each feature matrix was expanded along the new axis towards the model's input shape so that it would be compatible during training.

6. Preprocessing Validation and Test Sets: The same preprocessing was done for the validation and test sets as in the case of the training set; therefore, the same normalization and reduction of dimensionality were carried out. Their emotion labels were one-hot encoded into binary matrices for showing all the data uniformly.

The combination of these preprocessing steps enhanced the robustness of the training process, reduced the biases related to data, and prepared the dataset for efficient model learning and evaluation.

3.2 Model Building and Training

In this part, we describe the methodologies used for building and training the various machine learning models for speech emotion recognition. We employed multiple architectures, including CNN, LSTM, BiLSTM, CNN-LSTM, WaveNet, and DNN. Each model was trained on the extracted features to classify emotions from speech data.

3.2.1 Convolutional Neural Network (CNN)

CNNs are highly effective for identifying spatial patterns in data. In our project, we employed CNNs to extract localized features from audio signals.

The general structure of a CNN model is:

1. Applies convolution operations to the input data to extract features.

$$y = f(W * x + b)$$

where W is the weight matrix, $*$ denotes convolution, x is the input, b is the bias, and f is the activation function.

2. Pooling Layer: Decreases the dimensionality of the feature map while preserving key information.

$$y = \text{pool}(x)$$

where pool is the pooling operation (e.g., max pooling).

3. Fully Connected Layer: Combines features to make the final prediction.

$$y = f(Wx + b)$$

where W is the weight matrix, x is the input, b is the bias, and f is the activation function.

3.2.2 Long Short-Term Memory (LSTM)

LSTM networks, a form of RNN, can capture long-term dependences and are effectively applied to handle sequential information, like audio data.

The structure of an LSTM cell includes:

1. Forget Gate: Determines which information should be discarded in the cell state.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

where σ is the sigmoid function, W_f is the weight matrix, h_{t-1} is the previous hidden state, x_t is the current input, and b_f is the bias.

2. Input Gate: Manages the addition of new data to the cell state.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\widetilde{C_t} = tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \widetilde{C_t}$$

where i_t is the input gate, $\widetilde{C_t}$ is the candidate cell state, and C_t is the updated cell state.

3. Output Gate: Controls the release of information from the cell state.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \cdot tanh(C_t)$$

where o_t is the output gate and h_t is the hidden state.

3.2.3 BiLSTM (Bidirectional Long Short-Term Memory)

BiLSTM is a type of RNN that processes data in both forward and backward directions, capturing dependencies from both past and future contexts. This is particularly useful in tasks like speech emotion recognition, where context from both directions can be important.

The structure of a BiLSTM model is:

1. The BiLSTM computes forward hidden states $\underset{h_t}{\rightarrow}$ and backward hidden states $\underset{h_t}{\leftarrow}$:

$$\vec{h_t} = LSTM_{forward}(x_t, \vec{h_{t-1}})$$

$$\overleftarrow{h_t} = LSTM_{backward}(x_t, \overleftarrow{h_{t+1}})$$

2. The final output is a concatenation of these states:

$$h_t = [\vec{h_t}, \overleftarrow{h_t}]$$

3.2.4 CNN-LSTM Model

This model combines CNNs and LSTMs to capture both spatial and temporal features from the data.

The CNN layers extract spatial features, which are then fed into LSTM layers for temporal analysis:

$$\mathrm{CNN(X)} \rightarrow \mathrm{LSTM(features)}$$

3.2.5 WaveNet Model

WaveNet is a model created to generate raw audio by leveraging dilated causal convolutions, which effectively capture temporal dependencies over long periods. This method allows the network to expand its receptive field without increasing the number of parameters, thanks to the use of dilated convolutions. WaveNet is designed for audio synthesis but can be adapted for emotion recognition.

The dilated convolution operation is given by:

$$(f *_d x)(t) = \sum_{k=0}^{K-1} f(k) \cdot x(t - d \cdot k)$$

where d is the dilation factor, f is the filter, and x is the input signal.

3.2.6 Deep Neural Network (DNN)

DNNs are composed of multiple layers of neurons that process input data to generate output predictions. Each layer applies a linear transformation to the data, and a nonlinear activation afterwards.

The output of a neuron in a DNN layer is computed as:

$$y = \sigma(Wx + b)$$

where W is the weight matrix, x is the input vector, b is the bias, and σ is the activation function.

3.3 Model Evaluation and Comparison

By observing the training and testing accuracy and loss curves, we can determine the two best-performing models. Once the two best models are determined, we use the confusion matrix and classification report to analyze their performance in detail. The confusion matrix shows the prediction results for each category, helping us understand which categories the models perform well on and which need improvement. The classification report provides precision, recall, and F1-score for each category, as well as the overall accuracy. These metrics give us a comprehensive evaluation of the model's performance across different categories.

Classification report:

Precision:

$$= \frac{True\ Positive}{True\ Positive + False\ Positive}$$

Recall rate:

$$= \frac{True\ Positive}{True\ Positive + False\ Negative}$$

F1 score:

$$= 2 \cdot \frac{\text{Recall} \cdot \text{Precision}}{\text{Recall} + \text{Precision}}$$

4. Solution

The goal of this research is to construct an efficient speech emotion recognition system that can automatically detect the emotional information in audio. A deep learning-based emotion recognition model was designed and implemented for this purpose. This design includes data collection, feature extraction, data enhancement, model selection, and training, etc. Unlike traditional speech emotion recognition, we adopt several feature extraction methods and synthesis methods of different models to find the best combination scheme.

4.1 Data Preparation

We utilized four widely recognized emotional speech datasets: RAVDESS, CREMA-D, TESS, and SAVEE. These datasets provide a diverse range of emotional expressions, including emotions such as neutral, calm, happy, sad, angry, fearful, disgust, and surprise. We combined these datasets to form a comprehensive dataset for our experiments.

The data was first organized into a unified format. Each audio file was labeled with its corresponding emotion, and all file paths were consolidated into a single DataFrame. This allowed for efficient data manipulation and ensured consistency across the datasets.

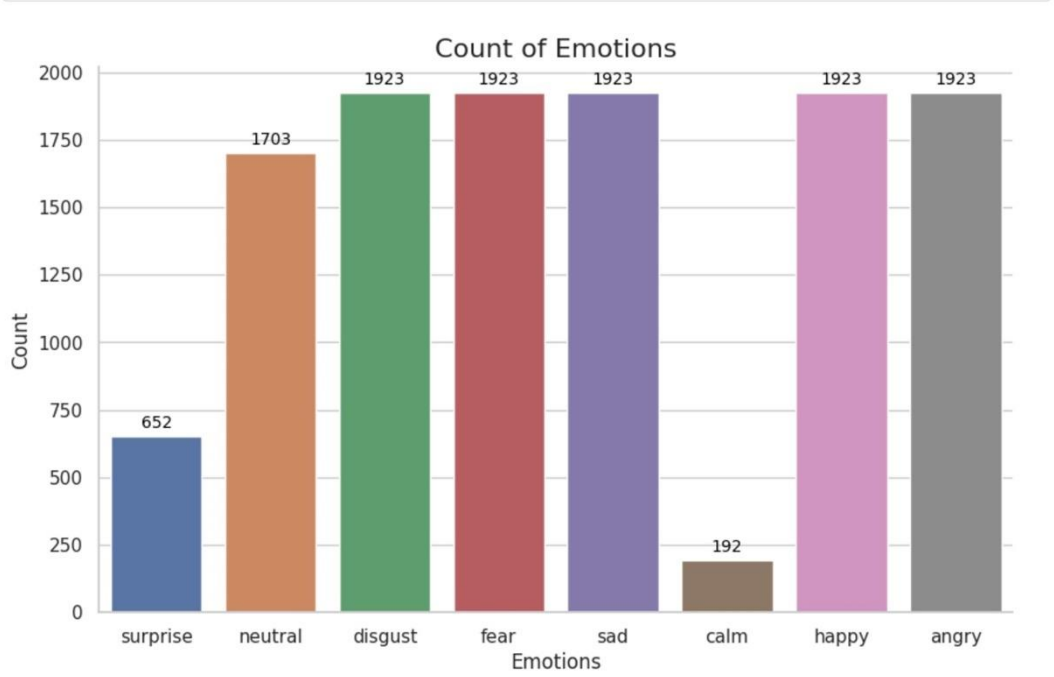


FIGURE 6. General SER System.

It can be seen that there are a total of 8 emotions in this integrated dataset, and emotional distribution in the original dataset is uneven, which may affect the overall prediction accuracy in the end. But this is not a big problem, we will solve it later.

To prepare the data for modeling, we performed a stratified split to divide the dataset into training, validation, and testing sets with a ratio of 70:15:15. Stratification

ensured that each subset had an equivalent distribution of emotion classes, which is crucial for balanced training and evaluation.

4.2 Data Augmentation

Data augmentation produces synthetic samples by applying light modifications to the original training dataset. For audio, techniques will be adding noise in the back ground, time-axis stretching or shifting of audio in time and pitch variation. These methods have been chosen since they provide good variability that may make the model more robust for these changes and enhance generalization capability. Below are the specific data augmentation methods and steps (taking any example and checking for techniques):

4.2.1 Raw data:

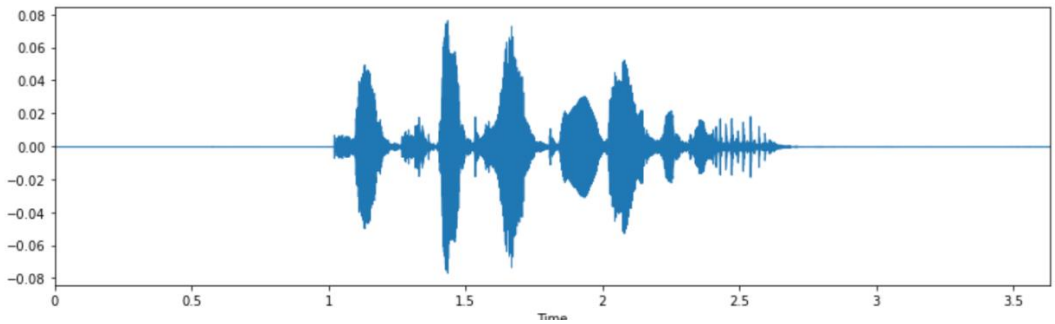


FIGURE 7. Waveplots of one raw data

4.2.2 Noise Injection:

Noise injection adds random noise to the original audio data. We use the np.random.uniform (noise_amp Calculated noise amplitude) and np.random.normal functions (Generates normally distributed noise of the same length as the data) from the librosa library to generate noise and add it to the audio data.

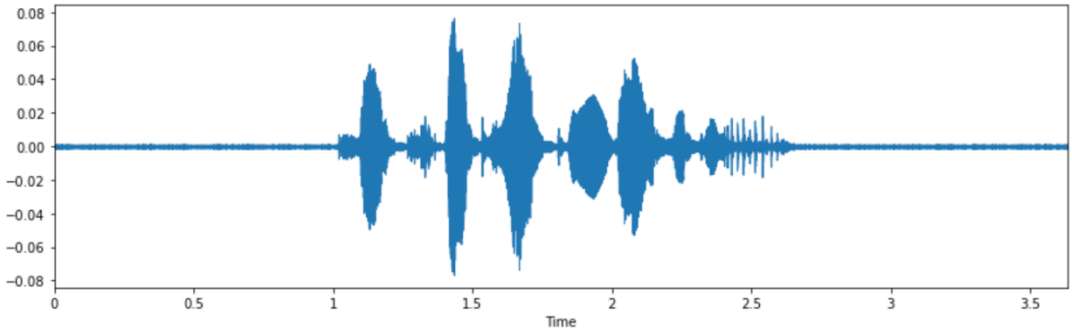


FIGURE 8. Waveplots of the raw data after Noise injection

We can see noise injection is a very good augmentation technique because of which we can assure our training model is not overfitted.

4.2.3 Time Stretching:

Time stretching simulates changes in speaking speed by altering the playback speed of the audio signal. We use the librosa.effects.time_stretch function to achieve

this (rate is the speed factor, less than 1 indicates slowing down, greater than 1 indicates speeding up).

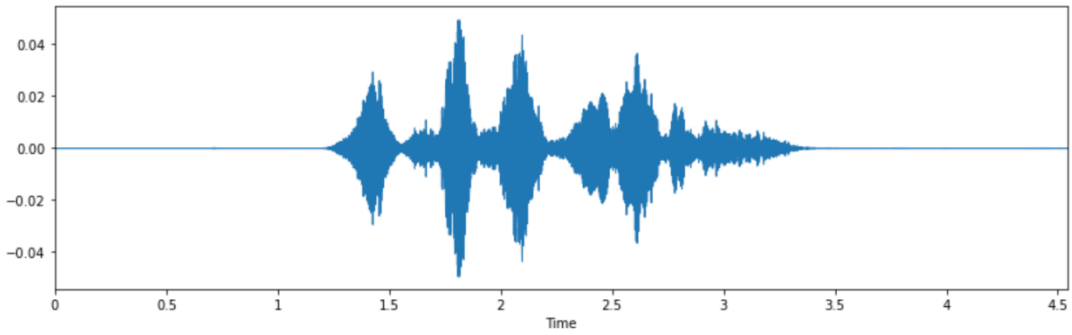


FIGURE 9. Waveplots of the raw data after Time stretching

4.2.4 Time Shifting:

Time shifting moves the audio signal along the time axis (shift_range Generates a random integer between -5000 and 5000, representing the range of translations). We use the np.roll function to achieve this.

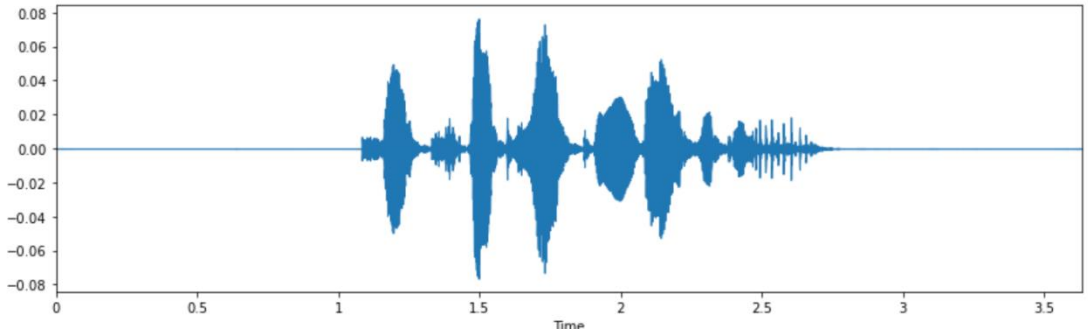


FIGURE 10. Waveplots of the raw data after Time shifting

4.2.5 Pitch Shifting:

Pitch shifting changes the pitch of the audio signal. We use the librosa.effects.pitch_shift function (To change the tone of the data, sampling_rate is the sampling rate and pitch_factor is the tone change factor. A value less than 1 indicates that the tone is lowered, while a value greater than 1 indicates that the tone is raised) to achieve this.

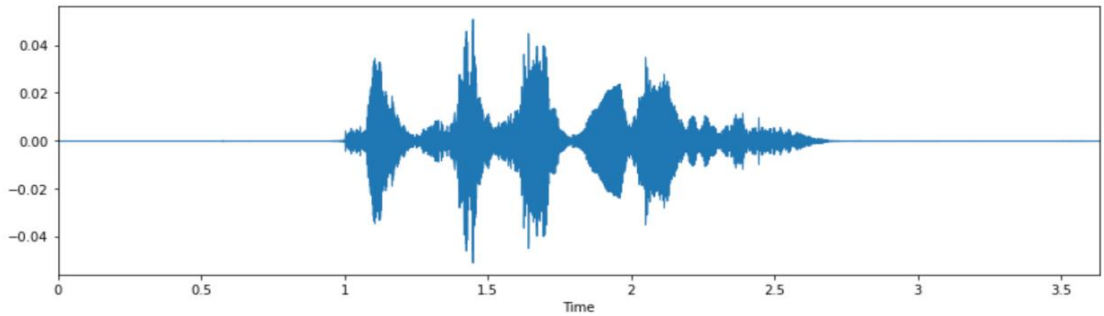


FIGURE 11. Waveplots of the raw data after Pitch shifting

From the above types of augmentation techniques we choose to use noise, stretching (ie. changing speed) and some pitching, since time shifting is not resulting very significant improvement.

4.3 Feature Extraction

The feature extraction process was conducted using three distinct methods, each chosen to represent various aspects of the audio data. These features are instrumental in enabling the model to effectively classify emotions, capturing both low-level and high-level emotional cues from speech.

4.3.1 Traditional Audio Features (Manually Designed Features)

Features like ZCR, Chroma_stft, MFCC, RMS Value, and Mel Spectrogram captured various essential aspects of the audio signal, including pitch, harmonic content, and frequency representation. These features provided a baseline to evaluate performance using traditional methods combined with machine learning models.

4.3.2 Wav2Vec 2.0

This self-supervised learning model by Meta extracts meaningful features from raw audio waveforms, capturing content and temporal dependencies directly without requiring labeled data during pre-training.

4.3.3 Emotion2Vec

This deep learning-based approach aims to extract features specifically optimized for emotion recognition, leveraging pre-trained models to capture high-dimensional emotional features.

4.4 Addressing Class Imbalance with SMOTE

The target was to balance all emotion classes to have an equal number of samples. Specifically, each class was adjusted to have 4,038 samples, matching the size of the largest class after augmentation. This balancing act was crucial to ensure that the model learned equally from all emotion categories.

Before applying SMOTE, emotion labels were encoded numerically using Label Encoding to meet the input requirements of the SMOTE algorithm. The sampling strategy was dynamically determined based on the original counts of each class to adjust only the minority classes.

After applying SMOTE, the dataset became balanced across all emotion categories, reducing the risk of model bias toward majority classes and improving overall classification performance.

4.5 Data Preprocessing for Neural Network Input Compatibility

To make the processed data set conform to the model inputs, the features were first standardized using StandardScaler. This standardization step needs to be applied to the processed training set, validation set, and test set. Furthermore, the transformed data is further reshaped by adding an extra dimension to prepare the input format required for the neural network model.

To ensure compatibility of output labels with classification loss functions, the OneHotEncoder method is used to one-hot encode the target labels. This method

converts labels into binary matrix format and facilitates multi-class classification tasks.

4.6 Model Selection and Training

We evaluated several machine learning models, including CNN, LSTM, BiLSTM, CNN-LSTM, WaveNet and DNN, to determine the most effective model for our SER system. Each of these models was trained and validated using different feature extraction methods to identify the combination that delivered the highest classification accuracy.

4.6.1 Initial Model Selection with Traditional Audio Features

Initially, we used manually designed audio features to train all the selected models. The DNN showed the best performance among all models when using traditional features, achieving a validation accuracy of 0.635 and a test accuracy of 0.623. The other models, including CNN and LSTM-based architectures, failed to achieve satisfactory accuracy with traditional features, all staying below 0.6. This led us to conclude that, while traditional features provide a reasonable baseline, they are insufficient for achieving high accuracy with more complex models.

4.6.2 Advanced Feature Extraction with Emotion2Vec and Wav2Vec 2.0

After the assessment of the traditional features, the focus was turned toward advanced methods, namely Emotion2Vec and Wav2Vec 2.0, to perform feature extraction. While combined with a DNN, the performance of Emotion2Vec was impressive; a validation accuracy as high as 80.5% and a test accuracy as high as 78.5% were found. The next best approach was from Wav2Vec 2.0, which attained 72.3% and 71.7% on validation and test accuracies, respectively, when combined with DNN. These results have proven the efficiency of advanced feature extraction methods driven by data in enhancing the performance of SER systems.

4.6.3 Final Model Architecture

Therefore, the final model was built on the comparative evaluation using a DNN in conjunction with Emotion2Vec features. The architecture of the network had multiple layers, consisting of fully connected layers with ReLU for the introduction of non-linearity and dropout layers to handle overfitting issues. It featured a softmax output layer, categorizing emotions into eight classes. Other parameters included categorical cross-entropy loss during training and the use of the Adam optimizer for efficient and stable convergence during optimization.

4.6.4 Some Details

The complete training pipeline was implemented using Python with libraries such as TensorFlow and scikit-learn. Feature extraction and augmentation were performed using Librosa, and SMOTE was applied using the imbalanced-learn package. Training and evaluation were carried out using Keras with GPU acceleration to expedite the process.

5. Validation

5.1 Model Evaluation

In fact, the final evaluation of the SER system was an overall performance assessment taken for the final DNN model based on the validation and test set. It involved several metric scores, viz., accuracy, precision, recall, and F1-score, that reflected the ability of the model to correctly classify various categories of emotional states.

The model was trained for 50 epochs while observing its training and validation accuracies for convergence and overfitting. ReduceLROnPlateau was used to reduce the learning rate on the appearance of a plateau in the validation loss, enabling even finer weight adjustments. The best model reached a validation accuracy of 0.805, proving it very good at generalizing on data it had not seen.

Result:

Model	Accuracy	Validation accuracy	Test accuracy
Tradition-fea-CNN	0.947	0.582	0.568
Tradition-fea-LSTM	0.671	0.550	0.550
Tradition-fea-BiLSTM	0.802	0.576	0.570
Tradition-fea-CNN-LSTM	0.805	0.604	0.568
Tradition-fea-WaveNet	0.752	0.584	0.563
Tradition-fea-DNN	0.761	0.635	0.623
Wav2vec2.0-DNN	0.827	0.723	0.717
Emotion2vec-DNN	0.908	0.805	0.785

TABLE 1: Waveplots of the raw data after Pitch shifting

Accuracy and loss curves were plotted to visualize the training process. The training and validation accuracy curves showed that our model did a good job.

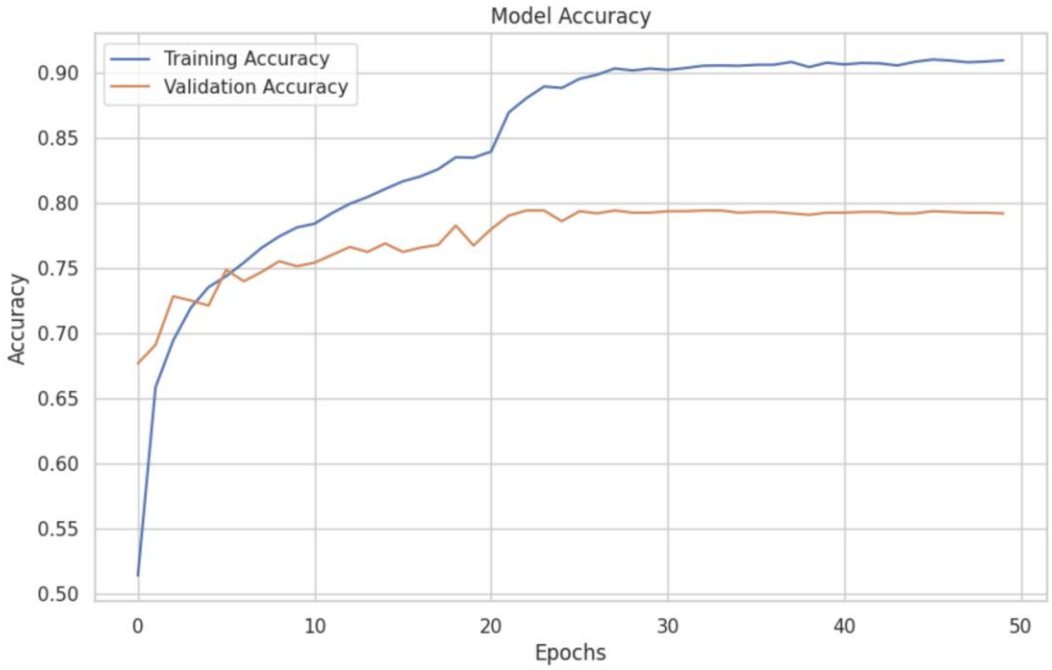


FIGURE 11: Accuracy varying with training epochs

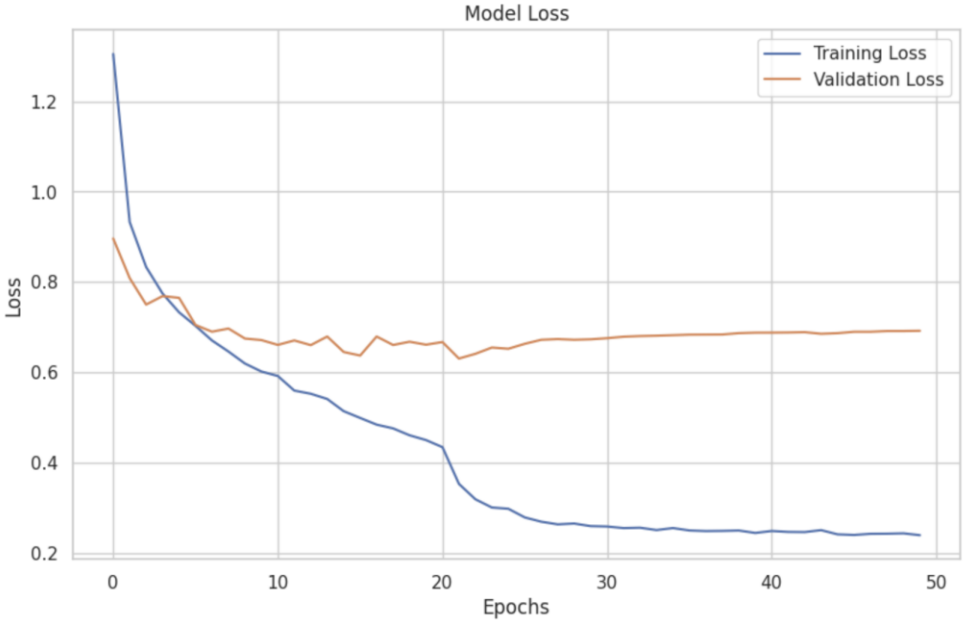


FIGURE 12: Loss varying with training epochs

5.2 Test Performance

5.2.1 Test Set Separated from the Original Dataset

Finally, the performance of the final DNN model was evaluated on the test dataset in conditions of a completely unseen dataset. The test accuracy was 0.785,

which was rather close to the validation accuracy, signifying that the model did not overfit much and was also generalizing quite well to new data.

5.2.2 Confusion Matrix and Classification Report

A confusion matrix was generated so as to present the model performance with respect to all categories of emotions. According to the confusion matrix, the model is performing well in recognizing most emotions; even though it has done best for "surprise", it is not good behind with respect to "fear" and "sad".

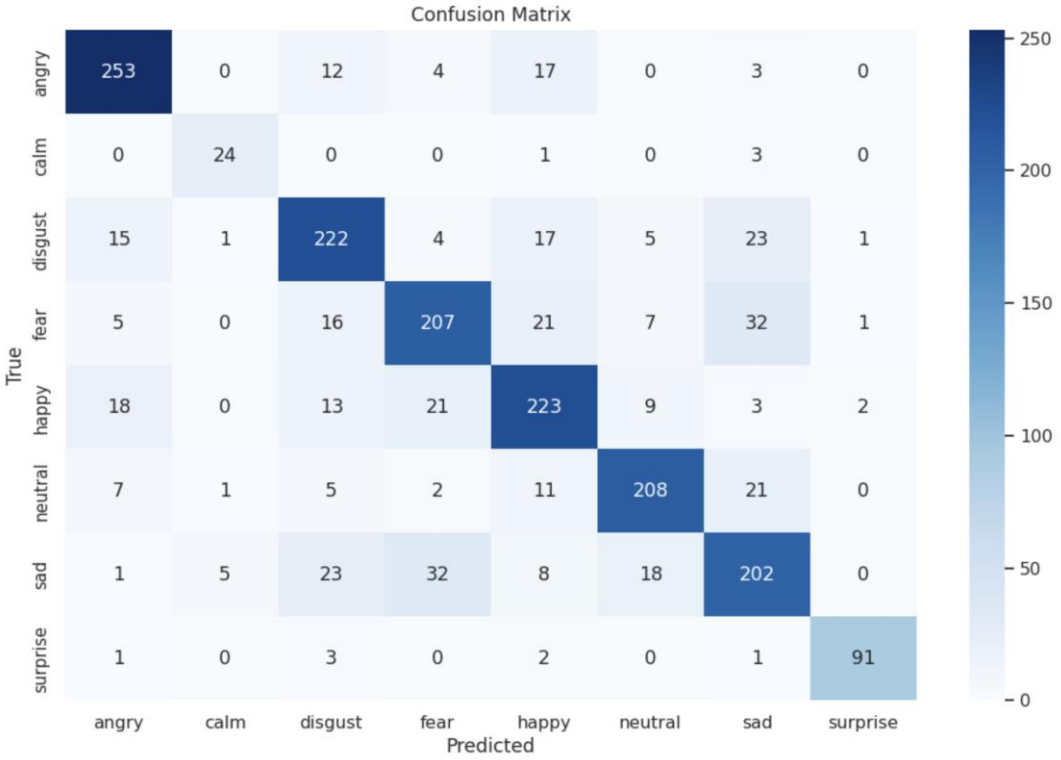


FIGURE 13: The Overall Confusion Matrix

The classification report provided detailed metrics for each emotion category:

- Precision ranged from 0.71 to 0.96, indicating that the model correctly identified emotions most of the time.
- Recall values showed that the model effectively detected emotions, with most categories achieving above 0.70 recall.
- F1-score, which combines precision and recall, confirmed the model's balanced performance, with an overall F1-score of 0.79.

```
Classification Report:
              precision    recall  f1-score   support

       angry       0.84      0.88      0.86       289
        calm       0.77      0.86      0.81        28
     disgust       0.76      0.77      0.76       288
        fear       0.77      0.72      0.74       289
       happy       0.74      0.77      0.76       289
     neutral       0.84      0.82      0.83       255
         sad       0.70      0.70      0.70       289
    surprise       0.96      0.93      0.94        98

    accuracy                           0.78      1825
   macro avg       0.80      0.80      0.80      1825
weighted avg       0.78      0.78      0.78      1825
```

FIGURE 14: The Overall Scores

5.2.3 Self-Recorded Test Set (Self_Test Dataset)

Along with this test dataset, we also used a self-recorded dataset—self_test, which includes 69 audio recorded by non-professional individuals of us. This is to simulate situations in the real world, where emotions are not always perfectly expressed. Performance on this dataset remained much lower, with a test accuracy of 44.93%. The classification report identified that some emotions, like "calm" and "surprise," are more subtle in terms of vocal expression and hence were a bit challenging for the model to classify correctly. It is expected that there should be this difference in performance, because non-professional recordings may evade emotional expressiveness compared to the recordings in the original dataset, since replicating the emotions with identical level and clarity may be rather difficult in the absence of professional guidance.

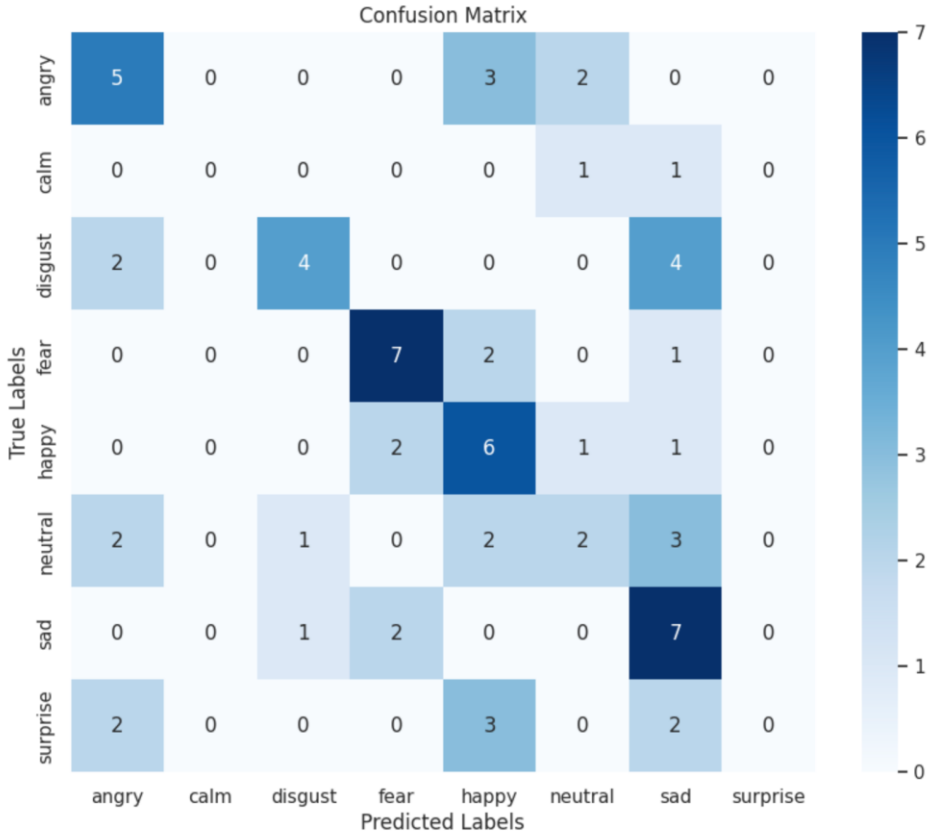


FIGURE 15: Confusion Matrix of Self-Recorded Test Set

测试集准确率: 44.93%
分类报告:

	precision	recall	f1-score	support
angry	0.45	0.50	0.48	10
calm	0.00	0.00	0.00	2
disgust	0.67	0.40	0.50	10
fear	0.64	0.70	0.67	10
happy	0.38	0.60	0.46	10
neutral	0.33	0.20	0.25	10
sad	0.37	0.70	0.48	10
surprise	0.00	0.00	0.00	7
accuracy			0.45	69
macro avg	0.35	0.39	0.35	69
weighted avg	0.41	0.45	0.41	69

FIGURE 16: Accuracy of Self-Recorded Test Set

5.3 Comparative Analysis of Features and Models

The comparative analysis of different feature extraction techniques and models underscored the superiority of Emotion2Vec combined with DNN. The performance

of Wav2Vec 2.0 was also noteworthy, achieving a test accuracy of 0.717, which, while lower than that of Emotion2Vec, demonstrated the effectiveness of self-supervised learning for feature extraction. Traditional features, in contrast, were unable to deliver competitive results, especially when applied to more complex models such as CNN and LSTM.

5.4 Limitations and Future Directions

While the model achieved high accuracy, there were some limitations. The model struggled with distinguishing between certain emotions, particularly those with similar vocal characteristics, such as "fear" and "sad." Additionally, the computational requirements for training Emotion2Vec and Wav2Vec 2.0 are significant, which may limit the practicality of deploying these models in real-time applications without adequate resources.

The self-recorded (self_test) dataset highlighted another limitation: the model's difficulty in accurately predicting emotions from non-professional recordings. The lower performance on this dataset suggests that the model is sensitive to the quality and intensity of emotional expression, which can vary in real-world scenarios.

6. Conclusion and Future Work

This paper focuses on a deep learning approach for Speech Emotion Recognition. The dataset compiled for this study was a combination of four datasets: CREMA-D, RAVDESS, SAVEE, and TESS. This dataset had a wide variation in the range of expressed emotions. To overcome problems such as class imbalance and limited variability, some of the used data augmentation techniques were noise addition, time stretching, pitch manipulation, among others. Synthetic Minority Over-sampling Technique was also used for maintaining dataset balance.

Different feature extraction methods were explored, such as traditional audio features and state-of-the-art deep learning representations including Emotion2Vec and Wav2Vec 2.0. Of the different ways being tested, the best performance was Emotion2Vec features coupled with a Deep Neural Network architecture, returning 80.5% validation accuracy and 78.5% test accuracy. This in turn speaks much about the importance and worth of features custom-designed for emotion recognition tasks.

The contributions of this work are threefold: it successfully integrated multiple datasets to improve the generalization ability of the model, employed recent feature extraction methods to capture complex emotional expressions, and developed a high-performance SER model by using DNN and Emotion2Vec features. Besides, several different models and feature extraction techniques were compared, and this comparison shed light on how to direct future studies in the SER domain.

However, some limitations have been identified: the model generalizes poorly to real-world scenarios, as evidenced by a low accuracy of 44.93% on the self-recorded test set. That would point toward sensitivity of the system to differences in recording quality and intensity of emotional expression common in non-professional settings. Furthermore, the more complex models-such as Emotion2Vec-are computationally intensive to train and thus limit usage for real-time applications.

Future work should be toward increasing robustness by considering more varied data with instances of spontaneous speech. This could allow models to generalize well to non-professional environments through transfer learning and domain adaptation. Advanced models, if optimized for computational complexity or using lightweight architectures, would aim at better feasibility for resource-constrained systems. Expanding emotional categories and considering multimodal data, including visual and physiological cues, would explain more about human emotions.

In general, the current research brings into view effective deep learning models combined with advanced feature extraction methods in SER. It forms the basis for further advances toward more versatile and efficient SER systems with very promising consequences in human-computer interaction and various practical applications.

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